

(a.) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

- From odds to probability: $p = \text{odds} / (1 + \text{odds})$
- From probability to odds: $\text{odds} = p / (1 - p)$

$$p = 0.37 / (1 + 0.37) = 0.37 / 1.37 \approx 0.2701$$

$$\text{odds} = 0.27 / (1 - 0.27) = 0.3699 \approx 0.37$$

(b.) Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?

$$\text{odds} = 0.16 / (1 - 0.16) = 0.16 / 0.84 \approx 0.1905$$

The odds she will default is 0.1905